

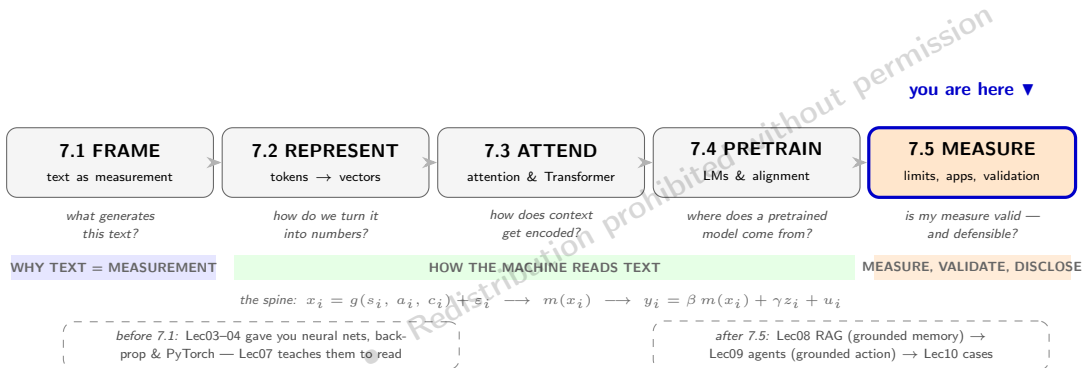
AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 7.5a: Limits of LLMs

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Lecture 7 in One Map



Route: five structural limits, each traced from its architectural root to its research implication (part 1/2 of MEASURE — what breaks); part 2/2 covers what works, and how to defend it.

Five Structural Limits

Architecture in, research implication out

Five Limits, Five Architectural Roots

None of these are bugs awaiting a patch — each follows from how the machine is built.

Limit	Architectural root	Research implication
1. Knowledge cutoff	knowledge frozen in weights; updating = retraining	provide the sources in the prompt; never assume “it knows the event”
2. Hallucination	max-likelihood over <i>form</i> , not truth	every citation and number is a draft until independently verified
3. Context window	attention cost $O(n^2)$	chunking severs cross-references; full filings may not fit
4. Lost in the middle	positional attention bias (U-curve)	extraction accuracy depends on <i>where</i> the passage sits — non-classical error
5. Prompt sensitivity & primacy	ICL: the prompt is the prior	freeze, version, and publish prompts; randomize option order

The next five frames take each limit in turn; the capstone shows them compounding into a 3.6× measurement gap across annotator models.

Limit 1 — The Knowledge Cutoff

- Every pretrained LLM is trained on a *static snapshot* of text. After the snapshot ends, the model learns nothing new.
- Each model has a **knowledge cutoff date**. After this date:
 - Events are invisible to the model
 - New papers, datasets, policy announcements simply don't exist as far as it is concerned
 - Asking “what happened in March 2026?” against a model trained through 2024 will at best yield “I don't know” — and at worst, a confident fabrication
- **Why not just retrain?**
 - Cost: \$10M–\$100M+ for a frontier model
 - Time: weeks to months
 - Energy and infrastructure constraints
 - Practically impossible to keep “current”
- This is not a bug — it is structural: the knowledge lives *in the weights*. Any solution that needs *fresh* information must work *around* the cutoff, not against it. **This is the core motivation for Lecture 8.**

Limit 2 — Hallucination

What it looks like

- Fabricated citations to papers that do not exist
- Made-up statistics that look plausible
- Non-existent laws, cases, article numbers
- Subtly wrong details in otherwise correct prose

Mitigations

- Require and verify citations externally
- Use retrieval (Lec08) to ground answers in real documents
- Use tools / code: numbers from a calculator
- Validate a sample by hand
- Set temperature $\tau = 0$ when measuring

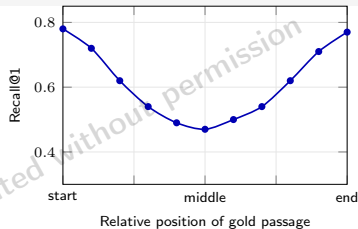
Why does it happen? The training objective (T4) maximizes the likelihood of *plausible-sounding* next tokens, not *true* ones. Where training data is sparse, the model fills the gap with the right *form* and invented *content*. Truth is a property of the world; plausibility is a property of language.

Limit 3 — Context Window and the $O(n^2)$ Wall

- Even modern models with 100,000+ token context cannot swallow arbitrarily large datasets:
 - A 200-page 10-K filing easily exceeds short context windows
 - Decades of FOMC transcripts cannot be loaded at once
 - Multi-firm or multi-country panels of textual data certainly cannot
- **Practical consequences:**
 - Long inputs must be *chunked*, which can break coherence and cross-reference — MD&A's reference to a risk factor stated 30 pages earlier is severed by the chunk boundary
 - Important information may sit outside the window altogether
 - Attention cost is $O(n^2)$ in the context length, so doubling n quadruples compute and memory
- Newer architectures (sparse attention, sliding-window attention, RoPE-extension, state-space hybrids) push the limit but do not eliminate it — and they trade away exact global attention (T3b).
- **Implication:** you cannot just dump “all the data” into a prompt. You need a way to *retrieve* the relevant pieces and feed them in — another argument for RAG.

Limit 4 — Lost in the Middle: Positional Saliency

- **Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni, Liang** (TACL 2024): when the relevant fact sits in the *middle* of a long prompt, recall drops 20–30 percentage points relative to its position at the start or end.
- **U-shaped recall curve.** Across closed (GPT-3.5/4) and open (LLaMA-2, MPT) LLMs, accuracy is high when the gold passage is near the start or the end of the context window and sags in the middle — a positional bias in attention, not random noise.
- **Two implications:**
 - *Stuffing the prompt* (concatenate as much as it fits) is **not a substitute for retrieval**: padding adds noise that buries the signal.
 - RAG (Lec08) must therefore *rank* chunks, not merely retrieve them. Top-ranked chunks should be placed at the prompt's ends.



Stylized U-shape from Liu et al. (2024); same qualitative pattern across model families.

Limit 5 — Prompt Sensitivity and the Primacy Effect

Generic prompt:

Classify this sentence:
"Inflation remains elevated and the Committee remains highly attentive to inflation risks."
Label:

Model output: *neutral* — the descriptive verbs ("remains") and the absence of a directional action word dominate.

Role-conditioned prompt:

You are a monetary economist. Is the following sentence hawkish, dovish, or neutral?
"Inflation remains elevated and the Committee remains highly attentive to inflation risks."
Label:

Model output: *hawkish* — "highly attentive to inflation risks" is read as a tightening signal once the prior is monetary policy.

- Both prompts pass the same sentence to the same model. The difference is the *prior* the role + label space imposes (cf. ICL as Bayesian inference, T4).
- **Brucks & Toubia (2025, PLOS ONE)**: primacy bias on balanced binary tasks — GPT-4 frequently picks the *first* option (a strong, robust primacy bias). Randomize option order; average across runs.
- **WEIRD bias & version drift**: training corpus is Western/English-leaning; closed-source models change silently between API calls (T4's "version = instrument").
- **Take-away**. Treat the prompt as part of the research design — freeze it, version it, report it. Robust analyses report results under multiple paraphrases.

Capstone: When the Limits Compound — YVP (2026)

Three frontier models, one frozen prompt, 18,797 O*NET tasks — and a $3.6\times$ gap.

- Yin, Vu & Persico (NBER WP 35110) re-run the standard AI-exposure rubric (Eloundou et al.) with three 2025–26 frontier models — everything else held fixed.
- Mean exposure differs $3.6\times$ across annotator models; pairwise agreement $\approx 57\%$, Cohen's κ as low as **0.36** (hand-coding publication standard: $\kappa \geq 0.75$).
- Downstream, the same DiD regression on each model's scores: individual-level coefficients vary $2.4\times$; the county-level estimate **flips sign** (significant negative $\rightarrow$ insignificant positive).

Model version = measurement instrument. Pin it, disclose it, and validate against a human gold standard — or the regression inherits whichever model you happened to call. T5b performs the full autopsy of this literature and builds the defensive workflow.

Discussion: which of the five limits drives the divergence? Mostly #5 — prompt/prior sensitivity compounded by each vendor's alignment choices (T4) — with #2 contributing at the rubric boundaries.

The Brain Metaphor

An LLM is like a very capable brain that finished school on a fixed date.

What the brain knows

- General language and reasoning
- Broad economics, math, programming
- Published papers up to its cutoff
- Common patterns, idioms, code recipes

What the brain does not know

- Anything that happened after the cutoff
- Your private FOMC corpus
- A specific firm's internal documents
- This morning's data release
- Your co-author's unpublished draft

Question: how do we let this brain access the information on the right *without* retraining it? (Lecture 8's opening question — and T5b's closing bridge.)

Bridge: Knowing the Limits, Use the Strengths

The audit is done; now the toolkit.

- Nothing in this deck says “don’t use LLMs.” It says *unvalidated* use is what fails — and each limit has a research-design counter: ground it, pin it, audit it, randomize it.
- **T5b**: what embeddings already measure (similarity, classification, networks); the case-study record from ECB Word2Prices to asset embeddings; the AI-exposure cautionary tale in full; and the six-step validation workflow — ending on the lecture’s thesis.

Arc: T1 FRAME → T2 REPRESENT → T3 ATTEND → T4 PRETRAIN → **T5 MEASURE**.